P510/1

Physics

Paper 1

Jul/Aug. 2026

3 Hours

WEST NILE SECONDARY SCHOOLS EXAMINATION COUNCIL

(WENSSEC)

Uganda Advanced Certificate f Education

PHYSICS

PAPER 1

TIME:3HOURS

INSTRUCTIONS TO CANDIDATES

This paper consists of TWO sections: Section A and Section B

Section A contains TWO compulsory items.

Section B is divided into TWO parts. Attempt ONE item from Part 1 and ONE item from Part 2.

All working must be clearly shown. Non-programmable scientific calculators may be used.

©2026 West Nile Secondary Schools Examination Council

Turn Over

CamScanner

SECTION A: COMPULSORY

ITEM 1

Uganda has made significant strides in space technology marked by the successful launch of its first Satellite – Pearl AfricaSat-1 on November 7, 2022. The project manager is intensifying mechanisms to develop more infrastructure to enable a second satellite launch before 2028. At Mpoma East station – Mukono, a satellite of mass 200 kg is to be launched in a parking orbit and the Project Manager needed to establish the height of the parking orbit, the mechanical energy of this satellite and get understanding of the effect of reducing the mechanical energy of the satellite on its safety. This information was necessary to establish safety precautions of the satellite in the orbit. The satellite is considered safe if its mechanical energy does not exceed -9.0×10^8 J. The project manager called one of the space Engineers in charge who had gone back to his nearby home to rush and return not later than 5 minutes to do the necessary calculations and deliver information. The engineer had parked his car on a road which was a straight stretch up to the station. He wants to start the engine of the car and give it a power of 22 kW for a driving force of 1.1 kN to be developed so as to uniformly accelerate for the first 1.2 km just before applying brakes for the remaining 0.5 km to decelerate uniformly to rest at the station. The project supervisor also tasked the engineer to explain how he can utilise some locally made materials to design a device that can produce a lift force with a mounted camera to capture aerial photos at the station.

Given:

Coefficient of friction on the used road = 0.025

Mass of the car = 2 tonnes

Acceleration due to gravity g = 9.81 m s^{-2}

Radius of the Earth $R_E = 6.4 \times 10^6$ m

Mass of Earth $M_E = 5.97 \times 10^{24}$ kg

Gravitational constant G = 6.67×10^{-11} Nm^2kg^{-2}

Task:

You have been selected to help:

(a) Analyse the data to provide the necessary information needed by the project manager and to establish if the engineer can reach in time to deliver it.

(b) Explain how the device needed by the supervisor can be made and able to serve the purpose.

CamScanner

Item 2

Dr. Jane, a medical physicist at Mulago National Referral Hospital in Kampala is reviewing the use of Iridium-192 (^{192}Ir) brachytherapy in treating cervical cancer patients. In this treatment, a radioactive source is placed inside the patient's body to deliver a focused dose of radiation. The hospital's radiation safety officer recorded that a freshly prepared sample of ^{192}Ir has an initial activity of

$A_0 = 3.70 \times 10^{11}$ Bq. It is known that ^{192}Ir decays with a half-life of 73.8 days. The safety officer needs to determine whether the source remains above the minimum therapeutic activity threshold of 1.50×10^{11} Bq after 120 days of use. Additionally, the physical principles underlying the interaction of emitted radiation with body tissue must be understood to advise on patient shielding and treatment room safety. The hospital is also evaluating whether locally available lead shielding is sufficient for the radiation energies involved.

A second patient, an accident victim was complaining of pain suspected to be due to fracture on the left leg. She directed a procedure for detailed information for the required treatment.

Task

(a) As a Physics student, identify and justify the physical principles operating in the scenario above, and evaluate the data provided to advise the radiation safety officer on whether the ^{192}Ir source remains therapeutically effective after 120 days of use, and what this implies for patient treatment scheduling.

(b) Identify the radiation detection device most suitable for monitoring personnel exposure in the hospital setting, evaluate how its operating principle makes it appropriate for detecting the type of radiation emitted by ^{192}Ir.

Also identify the equipment for getting the detailed information required for treatment of the second patient and explain how it is used for this purpose.

CamScanner

SECTION B
PART 1

*Attempt any **one** item from this part. The items carry equal scores.*

Item 3

Nurse Hope, in Koboko Hospital is simultaneously managing two critical tasks in the maternity ward. In the first task, she is monitoring a foetal heart using a device that emits very short bursts of high-frequency sound pulses at 5.0×10^6 Hz into the mother's abdomen and detects the echoes reflected back from the foetal heart wall. The foetal heart wall is moving toward the probe at 0.25 m s^{-1} through body tissue in which the speed of sound is 1,540 ms^{-1}. In the second task, Nurse Hope is ensuring the blood-bank refrigerator that stores donated blood at 4°C (277 K) is functioning correctly. The refrigerator extracts heat from its interior at 90 W and ejects it to the ward environment at 30°C (303 K). Its compressor works by rapidly squeezing a refrigerant gas from an initial volume of 9.0×10^{-3} m^3 at pressure 1.0×10^5 Pa to a final volume of 2.5×10^{-3} m^3 in an adiabatic process ($\gamma = 1.4$). Additionally, the neonatal incubator next to her has a polystyrene wall of thermal conductivity 0.033 $Wm^{-1}K^{-1}$, thickness 0.04 m, and surface area 1.8 m^2, maintained with an inner surface at 36°C and outer surface at 26°C. The hospital ventilation duct in the ward behaves as an open pipe of length 0.70 m and resonates at its fundamental frequency using air at 37°C in which the speed of sound is 352 ms^{-1}.

Relevant Relationship:

$COP_{CARNOT}(\text{refrigerator}) = T_C / (T_H - T_C)$

Task:

(a) As a Physics student, interrogate the physical phenomena operating across the various processes in Nurse Hope's scenario, identifying and explaining the principles at work and how they interconnect. Using the pre-calculated Carnot COP and the actual COP, critically evaluate the thermodynamic performance of the blood-bank refrigerator and advise the hospital biomedical team on whether the refrigerator's installed performance is satisfactory or whether intervention is necessary to safeguard stored blood quality.

(b) The device used to monitor the foetal heart in this scenario functions by directing very high frequency sound pulses into the body and detecting reflected echoes from internal structures. Explain in detail the physical principles by which this device operates and how the motion of the foetal heart wall produces a detectable change in the received signal that allows clinical information to be extracted.

CamScanner

Item 4

Mr. Kizito is a Senior Six Physics teacher at St. Mary's College Kisubi conducting a combined thermodynamics and waves practical lesson. He heats 0.50 kg of water (specific heat capacity 4,200 J kg^{-1} K^{-1}; specific latent heat of vaporisation 2.26×10^6 J kg^{-1}) from 25°C to boiling point (100°C) using an immersion heater rated at 1,500 W, then holds it boiling to evaporate 0.10 kg of water. This steam drives a small Stirling demonstration engine operating between 373 K (steam) and 303 K (room, 30°C), which absorbs 600 J of heat per cycle and delivers 112 J of work per cycle. On a separate bench, a student uses a resonance tube (a pipe open at both ends) of adjustable length to find the speed of sound. The tube resonates at its fundamental mode when its length is 0.33 m, using a tuning fork of frequency 520 Hz. Two tuning forks of frequencies 256 Hz and 260 Hz are also struck simultaneously nearby. A glass prism in the optics section of the lab separates white light into its constituent colours, demonstrating that the refractive index of the glass varies with wavelength; the refractive index for red light is 1.514 and for violet light is 1.532. The laboratory bench top (glass fibre, k = 0.04 $Wm^{-1}K^{-1}$, thickness 0.025 m, area 2.0 m^2) conducts heat from the hot apparatus to the room below, with the top surface at 60°C and the bottom surface at 30°C. Alongside the thermodynamics experiments, the biology section of the lab uses a device consisting of a short-focal-length objective lens (f_o = 5 mm) and a high-power eyepiece (fe = 25 mm) mounted coaxially in a tube to observe and measure the dimensions of micro-organisms on prepared slides.

Relevant Relationship:

$$\eta_C = 1 - T_C/T_H$$

Task;

(a) As a Physics student, identify and analyse the physical principles governing the various processes in Mr. Kizito's lesson scenario, connecting them to the relevant physics topics. Using the pre-calculated Carnot and actual efficiencies, critically evaluate the Stirling engine's thermodynamic performance, and using the bench conduction rate, comment on the thermal management implications of running such experiments indoors; advise Mr. Kizito on the overall safety and scientific integrity of the lesson setup.

(b) The biology section of the laboratory uses a device with a short-focal-length objective lens and a high-power eyepiece mounted coaxially. Explain the physical principles by which this device forms a highly magnified image of micro-organisms on prepared slides, and evaluate why a very short focal length objective is essential for achieving the magnifications required in biological microscopy.

CamScanner

PART 2

*Attempt any **one** item from this part. The items carry equal scores.*

Item 5

The Muni University Physics Society (MUPS) held its annual outreach open day, inviting S6 students from schools around West Nile to interact with Physics ahead of their UACE practicals. At one table, two capacitors of capacitances 4.0 μF and 6.0 μF were connected end-to-end and charged from a 240 V d.c. power pack, ready to briefly light up a row of LED bulbs spelling out "MUPS" once disconnected from the supply. At a second table, a group connected a known resistor and a rheostat to a battery and traced how the current changed as the potential difference across the resistor was varied, obtaining a straight line passing through the origin. A third group placed a plotting compass between two bar magnets arranged with similar poles facing each other and found that, no matter how the compass was nudged, it always settled motionless at one particular point along the line joining the magnets. Nearby, a fourth group connected a hot-wire instrument to the hall's lighting circuit and used it to convince visitors that, although the bulbs glowed steadily, the current actually reversed direction many times every second. As the outgoing MUPS president, you have been asked to write a short closing report for the incoming executive before you formally hand over office, who will also need to keep running the popular demonstration table where a large metal dome is rubbed until visitors can hear it crackle, producing a fine blue glow at the tip of a nearby grounded rod whenever the dome's charge grows large enough to leap across the gap.

Given:
$C_1 = 4.0\ \mu F$, $C_2 = 6.0\ \mu F$(connected in series)
Charging p.d. = 240 V
Maximum energy the LED signage driver can safely absorb = 50 mJ

Task:

(a) Justify the physics concepts evident at the first four tables described above, and evaluate whether the capacitor combination charged for the LED signage is safe for incoming members to operate, advising MUPS on any precaution necessary before handover.

(b) Justify how the dome at the popular demonstration table is able to produce the sparking effect observed, and assess its relevance to research or industrial practice beyond the demonstration.

CamScanner

Item 6

Kuluva Hospital in Arua District has long relied on its own small hydro-electric plant on the nearby waterfall to keep its wards and theatres running independent of the national grid, and the plant engineer is preparing for a planned increase in theatre activity. At the powerhouse, the engineer notes that the supply passes through a bank of capacitors installed years ago to correct the lagging current drawn by the laundry's washing machines and the workshop's drill press, that the theatre's anaesthetic monitor, like several other instruments, uses a small bridge of diodes inside its own power pack to convert the supply to direct current before it reaches the sensitive circuitry, and that the heavy aluminium feeder running from the powerhouse, through the hospital compound, to the main switchroom gets noticeably warm on busy days. The hospital administrator has also asked the engineer to explain, for a donor report, how the rotating coil at the powerhouse is able to generate the hospital's alternating supply as the falling water turns it.

Given:

Length of feeder cable $L = 350$ m

Cross-sectional area $A = 16$ mm^2

Resistivity of aluminium $\rho = 2.65 \times 10^{-8}\ \Omega$ m

Current at peak theatre load $I = 45$ A

Supply voltage = 415 V

Maximum acceptable voltage drop = 5% of supply voltage

Task:

(a) Justify the physics concepts evident in the engineer's inspection of the hospital's power system described above, and evaluate whether the feeder cable can deliver power to the switch room without an excessive voltage drop at peak theatre load, recommending what the hospital should do if it cannot.

(b) Assess how the rotating coil at the powerhouse generates the hospital's electricity as the water turns it.

END

CamScanner

Item 9 : Pg 1

• Period T of satellite in parking orbit = 24hr ✓M
= 24 × 3600 = 86,400 s ✓AN

Using $T^2 = \frac{4\pi^2 R^3}{GM}$ R = radius of orbit
M = mass of earth

$(86,400)^2 = \frac{4(3.14)^2 \times R^3}{(6.67\times10^{-11}) \times (5.97\times10^{24})}$ ✓AN

$R = 4.2241 \times 10^7$ m ✓AN

• Height of the parking orbit.

$h = (4.22\times10^7) - (6.4\times10^6) = 3.58\times10^7$ m ✓AN ✓AN ✓AN

• Mechanical energy of the satellite,

$M.E = -\frac{GmM}{R} = \frac{-(6.67\times10^{-11})(200)(5.97\times10^{24})}{(4.2241\times10^7)}$ ✓AN ✓AN

$= -1.887\times10^9$ J ✓AN

• When M.E reduces, satellite moves to a lower orbit. The velocity increases from $K.E = \frac{GmM}{2R}$. ✓ET ✓ET
friction increases. The satellite burns off. ✓ET

The M.E of the satellite -1.887×10^9 J is less than -9.0×10^8 J. ✓AN Since it will be safe if the energy does not exceed -9.0×10^8 J, the conclusion is that the satellite will be safe. ✓AN

Time taken by Engineer to reach.

1st stage of the motion is with constant acceleration a given by $T - f = ma$ f = friction, T = Tractive force
$= \mu R$

$\therefore (1.1\times10^3) - (0.025\times2000\times9.81) = 2000a$ ✓AN

$a = 0.30475$ ms^{-2}, ✓AN

$P = Tu_1$, u_1 = initial velocity,

$22\times10^3 = 11\times10^3 + u_1$ AN AN AN $u_1 = 20\ ms^{-1}$ ✓ AN

Using $s = ut_1 + \frac{1}{2}at_1^2$

$1\cdot2\times10^3 = 20t_1 + \frac{1}{2}(0\cdot30475)t_1^2$ ✓ AN

$t_1 = 30\cdot9\ s$ ✓ AN ✓ AN

Ex
3-4 — 5
1-2 — 3
0 — 0

2nd stage of Engineers motion.

$T = 0$ when brakes are applied.

AN
24-32 — 15

$F = ma_2$

$-f = ma_2$

16-23 — 10

$-0\cdot025\times2000\times9\cdot81 = 2000a_2$ ✓ AN

8 — 15 — 5

$a_2 = -0\cdot24525\ ms^{-2}$, ✓ AN

3 — 7 — 3

$v_2 = 0$, $u_2 = v_1$ where $v_1 = 20 + (0\cdot30475\times30\cdot9)$ AN

0-2 — 0

$= 29\cdot417\ ms^{-1}$ ✓ AN

$\therefore u_2 = 29\cdot417\ ms^{-1}$, $v_2 = u_2 + a_2t_2$

Hence $0 = 29\cdot417 + (-0\cdot24525)t_2$ ✓ AN

$t_2 = 119\cdot946\ s$ ✓ AN ✓ AN

Total time taken by the engineer to arrive AN

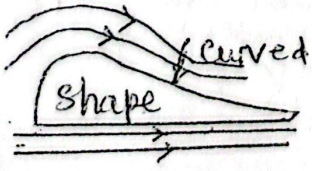
AP
3-4 — 05

$= 119\cdot946 + 30\cdot9 = 150\cdot846\ s = 2\cdot5141\ min$ ✓ AN ✓ AN ✓ AN

1-2 — 3
0 — 0

Conclusion: The engineer will reach earlier than 5 min. ✓ AN

(b)



Air at the top moves with higher velocity than lower part. ✓ AP

By Bernoulli's principle the pressure above is less than the one below. ✓ AP ✓ AP The pressure difference creates a lift force ✓ AP for the device.

Pg 2

ITEM 2: The principles involved are application of ... and X-rays ✓ EX

• Activity of Iridium-192, ^{192}Ir after 120 days

$$A = A_0 e^{-\lambda t}, \quad \lambda = \frac{0.693}{T_{\frac{1}{2}}} = \frac{0.693}{73.8 \times 24 \times 3600} \checkmark AN$$

$$= 3.70 \times 10^{11} \times e^{-\frac{0.693}{73.8 \times 24 \times 3600} \times (120 \times 24 \times 3600)} \checkmark AN$$

$$= 1.199 \times 10^{11} \text{ Bq} \checkmark AN$$

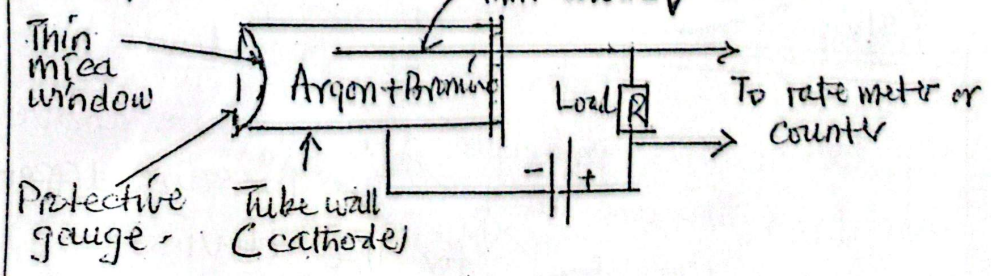
• Conclusion: This activity after 120 days will be less than the threshold therapeutic activity of 1.50×10^{11} Bq. Thus after 120 days, ^{192}Ir will not be effective for treatment. ✓ AN

• Advice: Treatment schedule should be less than 120 days. ✓ AN

• Principle involved in interaction of γ-rays with body cells:

γ-ray damages cells and DNA by knocking electrons out of atoms. ✓ EX The high energy process causes acute radiation sickness, ✓ EX organ failures and long term cancer risks ✓ EX when DNA code is changed by the γ-ray.

(b) The suitable device ✓ AP for detecting γ-rays is Geiger müller tube.



Pg4

The X-rays enter the tube through the thin mica window. They ionise the argon gas molecules producing positive and negative ions. The electrons travel to the anode and cause more ionisation by repeated collision resulting in to gas amplification. When they reach the anode, discharge occurs producing current and voltage pulse across R. The voltage pulse is amplified and fed in to the counter or ratemeter. The current pulse is proportional to extend of ionisation. Bromine gas is used as quenching agent.

EX-04
AN-08
AP-20

EX
3-4-5
1-2-3
0-0

The equipment for detecting fracture is X-ray tube. X-ray beam from the tube is passed through the part of the body being examined. Dense materials eg bones absorb most of the radiation. Less dense tissues eg skin, muscles, empty space of bone crack, allow the X-ray to pass through. The X-ray that passes through hits a photographic plate (film) on the other side. Areas where radiation was absorbed appear bright white (healthy bone) while areas where radiation passed through easily appear dark or black (the fracture).

AN
6-8-5
4-5-3
1-3-1
0-0

AP
15-20-15
10-14-10
5-9-5
2-4-3
0-1-0

ITEM 3:

The processes in the scinario are:

1. Doppler effect ✓ EX to monitor fetal heart beat.

The change in frequency (apparent frequency) of the ultrasonic sound wave $f' = \left(\frac{V_s - V_o}{V_w - V_o}\right) f = \left(\frac{0.25 - 0}{1,540 - 0}\right) \times 5 \times 10^6$ ✓ AN

$= 8.117 \times 10^2$ Hz ✓ AN

f - frequency of Ultrasonic sound
V_s - Velocity of source ✓ AN
V_o - Velocity of observer
V_w - velocity of ware.

2. Reversible isothermal and adiabatic processes in refrigerator.

The refrigerator removes heat from the cold Reservoir (interior) and discards it to hot reservoir (environment) and work is done on it.

Actual coefficient of performance, $COP = \frac{T_C}{T_H - T_C}$

$T_C = 4 + 273 = 277$ K ✓ AN, $T_H = 30 + 273 = 303$ K ✓ AN.

$COP = \frac{277}{303 - 277}$ ✓ AN $= 10.6$ ✓ AN

For the adiabatic process, $P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$

$(1.0 \times 10^5)(9.0 \times 10^{-3})^{1.4} = P_2 (2.5 \times 10^{-3})^{1.4}$ ✓ AN

$P_2 = 6.009 \times 10^5$ Pa. ✓ AN

Work done $W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{(9.0 \times 10^{-3})(1.0 \times 10^5) - (6.009 \times 10^5)(2.5 \times 10^{-3})}{1.4 - 1}$ ✓ AN ✓ AN

$= -1505.75$ J ✓ AN

Calculated $COP = \frac{Q_c}{W} = \frac{90}{1505.75}$ ✓ AN ≈ 0.06 ✓ AN

Actual COP > calculated COP. Refrigerator performance is satisfactory. ✓ AN

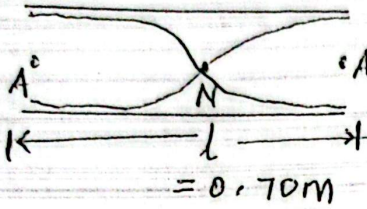
• Heat transfer; thermal conduction (ET) in neonatal incubator.

Rate of heat flow in the polystyrene layer (wall) of the incubator, $\frac{dQ}{dt} = \frac{KA(\theta_2 - \theta_1)}{l} = \frac{0.033 \times 1.8 \times (36-26)}{0.04}$ (AN)

$= 14.85\ Js^{-1}$ (AN)

• Stationary wave in open pipe (AN) applied in the ventilation duct.

EX-02
AN-25
AP-05



• A fundamental mode.

The distance between the anti-nodes, A and A, $\overline{AA} = \frac{1}{2}\lambda$ (AN) λ = wavelength of sound.

$\therefore \frac{1}{2}\lambda = l \Rightarrow \lambda = 2l$. $v = \lambda f \Rightarrow v = 2lf$

$v = 352\ ms^{-1}$ $\therefore$ $352 = 2 \times 0.70 f$, $f = 251.43 Hz$ (AN)

Fundamental frequency $f = 251.43 Hz$. (AN)

The device emits ultrasonic (AP) sound waves of frequency f, (High frequency).
The waves bounce off (are reflected) by the moving fetal heart. (AP)
The moving blood and walls alter (AP) the frequency of the ultrasonic sound.
The machine detects the frequency shift (AP); apparent frequency f'.
The shift (AP) becomes an audible heart beat sound.

ITEM 4;

The physics concepts in the scenario are:

- Specific heat capacity and latent heats. Ex

Heat supplied to heat water from 25°C to 100°C and evaporate 0.10 kg of the water

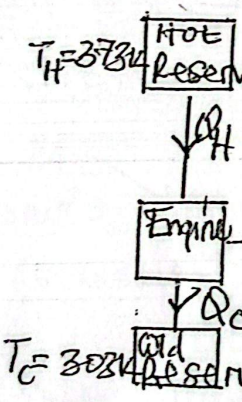
$= mc\Delta\theta + mL_V$ AN

$= 0.50 \times 4200 \times (100 - 25) + (0.10 \times 2.26 \times 10^6)$ AN AN

$= 157,500 + 2.26 \times 10^5$ AN

$= \underline{383,500\ J}$ AN

- Application of reversible isothermal and adiabatic process (Carnot cycle), in the sub-topic, changes with gases, in heat Engines. Ex



$Q_H = W + |Q_C|$ AN

$600 = 112 + |Q_C|$, AN $|Q_C| = 488\ J$ per cycle. AN AN

The maximum efficiency $\eta = \frac{W}{Q_H} \times 100 = \left(1 - \frac{T_C}{T_H}\right) \times 100$ AN

$= \left(1 - \frac{303}{373}\right) \times 100 = 18.7\%$ AN

- Thermal conduction (Ex) in the topic, heat transfer. (Ex)

Conduction by the bench top made of glass fibre.

Rate of heat flow through the fibre;

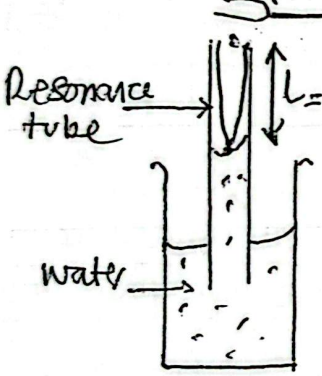
$$\frac{Q}{t} = KA\frac{(\theta_2 - \theta_1)}{L} = \frac{0.04 \times 2.0 (60-30)}{0.025} = 240\,J\,s^{-1}$$ (AN) (AN) (AN)

This is less than heat released (AN) by the engine to the cold reservoir. If the engine is run inside a room, the temperature inside will rise. (AN)

- Application of light (Ex) in optical instruments (Ex) particularly compound microscope. (Ex)

The objective of compound microscope has short focal length, $f_o = 5mm = 5 \times 10^{-3} m$

- Measurement of speed of ~~light~~ sound (Ex) using resonance tube (Ex) under the topic stationary waves. (Ex)



For Fundamental mode,

$$\frac{1}{4}\lambda = l \;,\; \lambda = 4l$$

$$v = \lambda f \;,\; v = 4lf = 4 \times 0.33 \times 520$$ (AN)

$$= 686.4\,ms^{-1}$$ (AN) (AN)

- When two tuning forks of nearly same frequency $f_1 = 256Hz$ and $f_2 = 260Hz$ are sounded together, Beats (Ex) are produced. (AN)

Beat frequency $f = f_2 - f_1 = 260 - 256 = 4Hz$ (AN) (AN) (AN)

- Dispersion of light by prism (Ex). In air/vaccum (Ex) all colours travel with same speed. In glass (other media), they travel with different (Ex) speeds.

Red has fastest speed and ~~Violet~~ Violet the least. Ex

Refraction is due to change in speed. Ex

Deviation of red light, $d_r = (n_r - 1)A = (1.514 - 1)A$

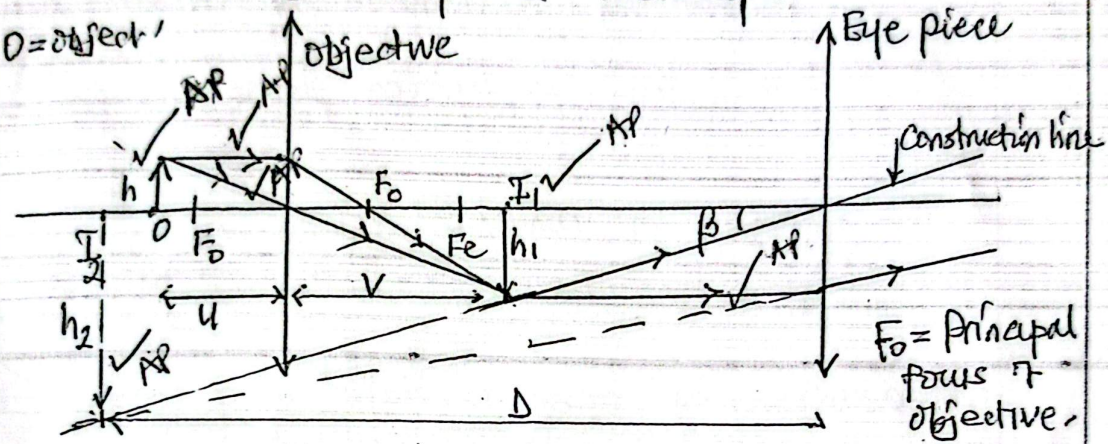
$= 0.514A$

Justification

Deviation of Violet light, $d_v = (1.532 - 1) = 0.532A$

Where A = angle of prism,

(b) The device is compound microscope.



In normal adjustment, final image I_2 is at least distance distinct vision D from eye piece. Object O is placed slightly beyond F_o. The object forms the inverted image I_1 in the microscope tube. Eye piece is adjusted so that, large Virtual image is formed by it at I_2. I_1 is nearer to the eye piece than Fe.

Angular magnification $M = \frac{\beta}{\alpha}$, β = angle subtended by I_2 (Aided eye)

α = angle subtended by object in unaided eye (Unaide eye)

$\beta = \frac{h_2}{D}$, $\alpha = \frac{h}{D}$ $\therefore M = \frac{h_2}{D} \times \frac{D}{h} = \frac{h_2}{h}$

But $\frac{h_2}{h} = \frac{h_2}{h_1} \times \frac{h_1}{h} \Rightarrow \frac{h_2}{h_1}$ = linear magnification of "object" at I_1 produced by eye piece.

$\Rightarrow \frac{h_2}{h_1} = \frac{V}{f_e} - 1 = -\frac{D}{f_e} - 1 = -\left(\frac{D}{f_e} + 1\right)$

$\frac{h_1}{h}$ = linear magnification of object by objective lens.

$\therefore \frac{h_1}{h} = \left(\frac{V}{f_o} - 1\right)$ $\therefore M = -\left(\frac{D}{f_e} + 1\right)\left(\frac{V}{f_o} - 1\right)$

M

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{V}{f} = M +$

$M = \frac{V}{f}$

- The focal length of the objective is very short because shorter focal length bends light more sharply to produce large, clear and highly magnified image of tiny objects.

- Magnifying power depends on focal length given by the equation,

$$M = \left(\frac{V}{f} - 1\right)$$, decreasing f increases magnification.

- Short focal length also improves the ability of the lens to separate close points (better resolution).

- It gathers more light from objects and shows sharp details in cells.

Ex - 17
AN - 20
AP - 27

X	AN	AP
5 - 17 - 5	15 - 20 - 10	20 - 27 - 10
- 12 - 3	10 - 14 - 8	14 - 19 - 8
- 7 - 1	5 - 9 - 5	7 - 13 - 5
- 3 - 0	2 - 4 - 3	5 - 6 - 3
	0 - 1 - 0	0 - 2 - 0

ITEMS:

The physics concepts ~~are~~ in the scinario are;

Table1:

CK-06 Combination of capacitors; series one and energy stored in a capacitor. EX

AN-07 Total capacitance $C = \frac{4\times6}{4+6} = 2.4\mu F$ AN AN AN

Energy stored in the 2.4μf capacitor Combination/

$E = \frac{1}{2}CV^2 = \frac{1}{2}\times(2.4\times10^{-6})(240)^2 = 0.0691\,J$ AN AN

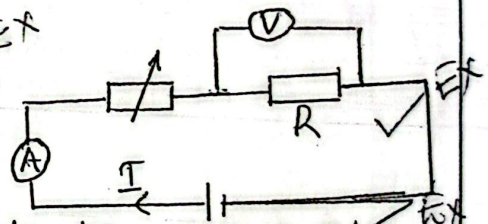
The maximum the LED can safely absorb = 50mJ

$= 50\times10^{-3}\,J = 0.050\,J$

Conclusion: The capacitor combination is not safe for the LED since the energy it stores 0.0691J is more than the maximum the LED can safely absorb. AN AN

Table 2:

1. Verification of ohms law. EX

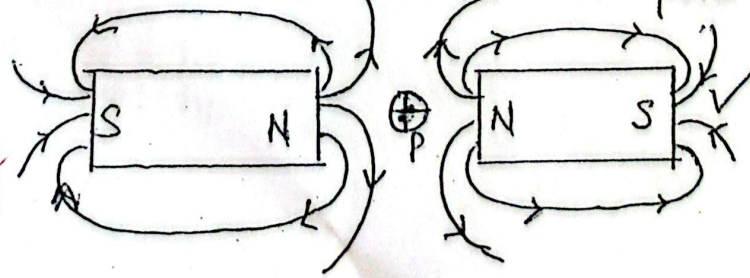


EX-04 The p.d across R is proportional to the current I. EX

$V \propto I \quad V = IR$ EX. Graph of V against I is a straight line

Table3:

1. Magnetic field between two like poles of a magnet. EX



EX-07 EX EX

P = Neutral point

If both are N-poles, the field lines point outwards away from each pole. If both are S-poles, the field lines point inwards towards each pole.

ITEMS:

The physics concepts ~~are~~ in the scinario are;

Table1:

Combination of capacitors; series one and energy stored in a capacitor.

Total capacitance $C = \frac{4 \times 6}{4+6} = 2.4\mu F$

Energy stored in the 2.4μF capacitor Combination/

$E = \frac{1}{2}CV^2 = \frac{1}{2} \times (2.4 \times 10^{-6}) / (240)^2 = 0.0691 J$

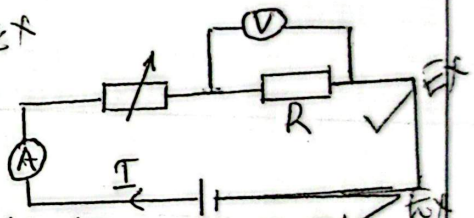
The maximum the LED can safely absorb = 50mJ

$= 50 \times 10^{-3} J = 0.050 J$

Conclusion: The capacitor combination is not safe for the LED since the energy it stores 0.0691J is more than the maximum the LED can safely absorb.

Table 2:

1. Verification of ohms law.

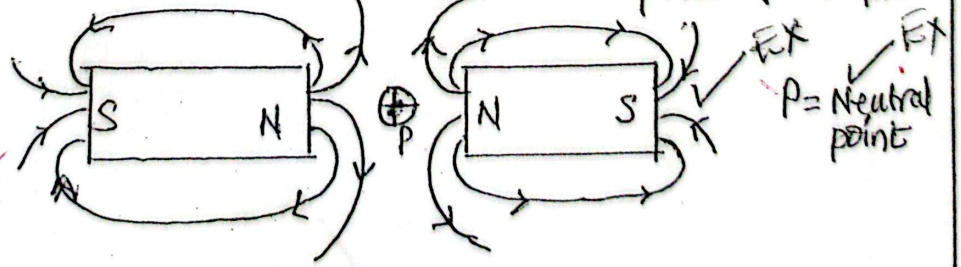


The p.d across R is proportional to the current I.

$V \propto I \quad V = IR$. Graph of V against I is a straight line

Table3:

- Magnetic field between two like poles of a magnet.

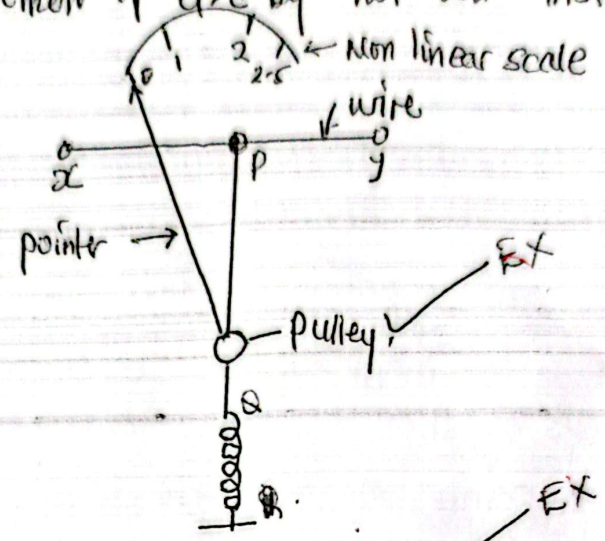


- If both are N-poles, the field lines point outwards away from each pole. If both are S-poles, the field lines point inwards towards each pole.

Field lines repel between the magnets. This creates a spot between the magnets with no field lines, zero magnetic field strength. This spot is called Neutral point. Compass placed at this point settles motionless.

Table 4

Is measurement of a.c by hot wire instrument



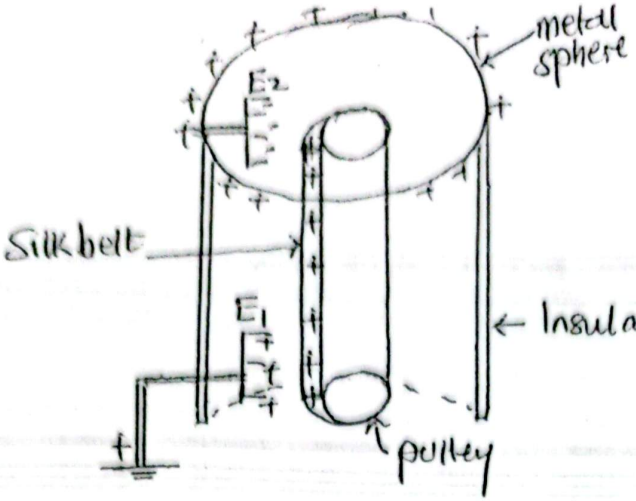
EX08

Current flows through XY, it heats it. This causes the wire to expand and sag. The sag is transmitted through PQ which passes round the pulley to which pointer is attached. The pulley rotates the pointer which deflects along the non-linear scale. The deflection is proportional to the average rate at which heat is developed in the wire XY. It is thus proportional to the root mean square current.

The pointer of the hotwire meter does not oscillate with the a.c because the wire responds to the average heating effect I^2R. Due to thermal inertia, the wire can not rapidly heat up and cool down at frequency of a.c. It stays at steady average temperature and shows stable root mean square value, I_{rms}.

shifted.

(b)



E_1, E_2 are electrodes

At sharp point of E_1, there is high electric field intensity. Point action takes place near the sharp point. Positive and negative ions are produced. The positive ions are repelled to the silk belt. The silk belt rises with them as it moves round the pulleys. When the +ve charges reach near the sharp point of electrode E_2, they induce -ve charges at sharp points of E_2 and +ve charges on the sphere. Point action also occurs near the sharp points of E_2. -ve ions produced are repelled to the silk belt and neutralize the +ve charges on the silk belt before it passes over the upper pulley so that the processes can be repeated. This way, there is continuous accumulation of charges on the sphere and produce very high p.d. This high p.d ionises air around. +ve and -ve ions are produced. Resulting in to visible electric sparks.

Application of the high p.d

- The high p.d is used in x-ray tube, to provide accelerating voltage.
- It is used for particle acceleration. The massive p.d creates strong electric field used to accelerate ions and protons at high speeds for nuclear physics and medical research.

Item 6: Physics concepts in the scenario are;

- Generation of a.c from hydroelectric plant.
- Conduction of a.c by capacitor to correct lagging.
- D.C power dispated in the aluminium cable. This causes heating of the cable

Power loss in the cable $= I^2R$

But $R = \frac{\rho l}{A} = \frac{(2.65 \times 10^{-8})(350)}{(16 \times 10^{-6})} = 0.580\ \Omega$

Power loss $= (45)^2 \times 0.580 = 1,174.5\ W$

Voltage drop $= IR = 45 \times 0.580 = 26.1\ V$

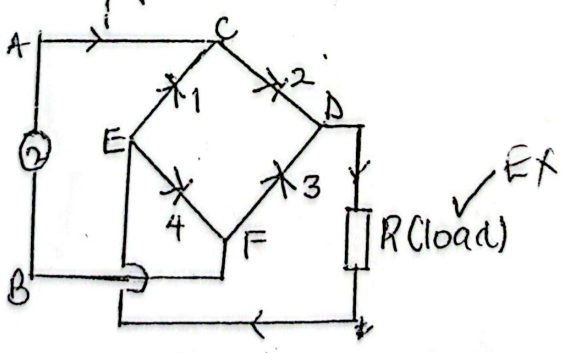
% of voltage drop $= \frac{26.1}{415} \times 100 = 6.289\%$

Since % of voltage is more than maximum 5% of 415 acceptable drop = 5%, the aluminium feeder cable is not suitable.

Recomendation: Use cable of larger cross-sectional area and thus lower resistance.

- Full wave rectification of a.c to d.c by 4 diodes.

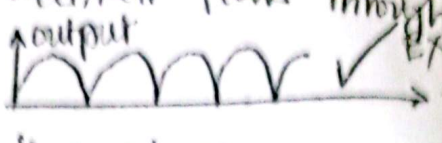
1, 2, 3, 4 – Diodes



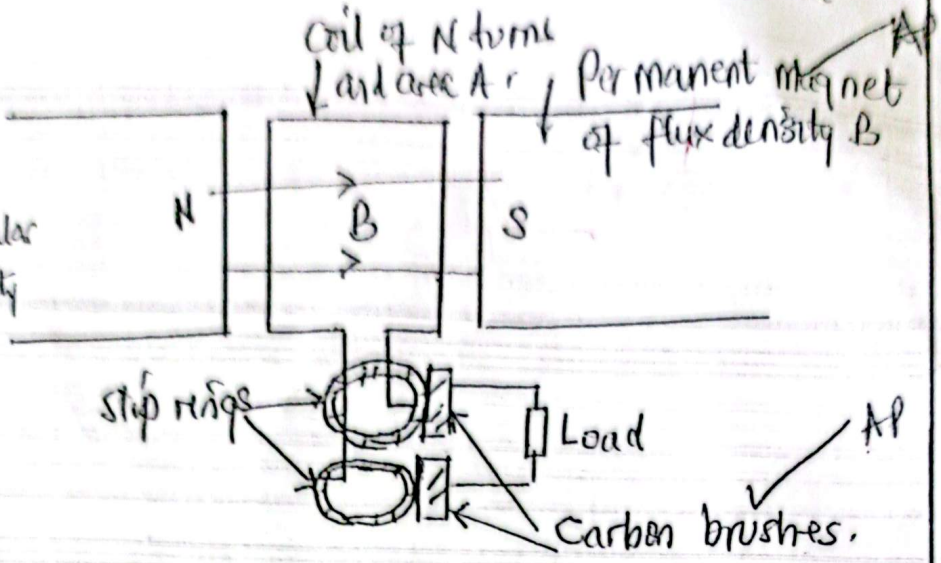
When A is positive, current flows from source to A, to C, through diode 2, to D, the load, to E, through diode 4 to F to B and back to source.

When B is positive, current flows from source to B, to F, through diode 3, to D, through R, to E through diode 1, to C, to A, back to source.

In both half cycles the current flows through R in the same direction.



(b)



The water fall, rotates the coil between the magnets. The magnetic flux $\phi = NAB\cos\omega t$ linking the coil changes as the coil rotates. This induces emf $= -NAB\omega\sin\omega t$ and current $i = -\frac{NAB\omega\sin\omega t}{R} = i_0\sin\omega t$,

$i_0 = \frac{NAB\omega}{R}$, R = resistance of the coil. The a.c flows through the load.

AR A.G

MARKING GUIDE

UACE PHYSICS 2026

PS101/1

CamScanner